

Fatty acid-induced inflammation and insulin resistance in skeletal muscle and liver.

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Plasma free fatty acid (FFA) levels are elevated in obesity. FFA, by causing insulin resistance in muscle, liver, and endothelial cells, contributes to the development of type 2 diabetes mellitus (T2DM), hypertension, dyslipidemia, and nonalcoholic fatty liver disease (NAFLD). The mechanism through which FFA induces insulin resistance involves intramyocellular and intrahepatocellular accumulation of triglycerides and diacylglycerol, activation of several serine/threonine kinases, reduction in tyrosine phosphorylation of the insulin receptor substrate (IRS)-1/2, and impairment of the IRS/phosphatidylinositol 3-kinase pathway of insulin signaling. FFA also produces low-grade inflammation in skeletal muscle and liver through activation of nuclear factor-kappaB, resulting in release of several proinflammatory and proatherogenic cytokines. Thus, elevated FFA levels (due to obesity or to high-fat feeding) cause insulin resistance in skeletal muscle and liver, which contributes to the development of T2DM, and produce low-grade inflammation, which contributes to the development of atherosclerotic vascular diseases and NAFLD.